

A Comparison of Asserted and Authoritative Data for the Haiti Earthquake of
January 2010

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Introduction and Background

The quality of geospatial data has been an important consideration since the birth of maps and geographic information systems (GIS) particularly due to the role of maps and spatial data in the decision making process (Chrisman 1991 and Foody 2003). While many people instinctively think of positional accuracy there are actually many components of geospatial accuracy. These components are lineage, positional accuracy, attribute accuracy, logical consistency, completeness, semantic accuracy, usage, purpose and constraints, and temporal quality (Van Oort 2006). Researchers have been developing tools and techniques to measure these forms of accuracy for years. While some of the tools and techniques can be quite complicated, the process has been helped historically by the rather limited ability to create geospatial data. Government agencies and private companies often relied on published standards to make sure that their data were of a sufficient quality. These data sources have become known as authoritative because of their understood inherent quality. Recently, however, the ability to create geospatial data has been opened to the masses. With the dropping costs for global positioning equipment, wireless internet access, computer hardware, and mapping software, many people are able to create geospatial data without any formal training. This data is referred to as asserted or volunteered in the literature (Goodchild 2007). This has led to an explosion in geospatial data volume in a relatively short time frame. Many GIS users and researchers have been quick to condemn this data produced by amateurs because the creators of the data are presumed to lack the necessary knowledge to properly collect data. Researchers in the United Kingdom, Germany, and France have compared data from the Open Street Map project (<http://www.openstreetmap.org>), a popular volunteer online mapping website, with data from authoritative sources like government agencies and in-car navigation companies. While the researchers certainly found room for improvement, in many instances the asserted, or volunteered, data was equal to the authoritative (Girres and Touya 2010, Haklay 2010, Haklay et al. 2010, Haklay and Ellul 2010, and Zielstra and Zipf 2010).

In the last several years volunteered geographic information (VGI) has been used for more time critical situations like natural and human caused disasters, in addition to more traditional applications like mapping roads. A web application named Ushahidi was first developed in Kenya in 2008 to help identify post election violence during media black outs (Okolloh 2008). Since that time the service has been used to map damage from earthquakes, floods, tsunamis, and civil unrest in the Middle East. While members of the public and relief agencies contribute to and use this volunteered data, there are still many traditional relief agencies who are leery of the data because of its uncertainty. Up to this point, however, geospatial researchers have not conducted any studies to either confirm or allay these fears.

The purpose of this project is to assess the positional accuracy, completeness, and temporal accuracy of data produced by volunteers during the Haiti earthquake of January 2010. While the techniques used in this project are not sufficient to make

broad generalizations about the reliability of volunteered geospatial information in time critical, disaster situations, it should help to outline potential methods to further pursue this topic.

Methods

This project compares different aspects of geospatial data quality in order to better understand the strengths and limitations of both VGI datasets and authoritative datasets in time critical, disaster situations. One method commonly used to compare data sets is to assume that one data set is of higher quality in one or more component of quality than another and to assess the weaker data set against the stronger (Goodchild and Hunter 1997). While this method works well when comparing Open Street Map data to Ordinance Survey data, it does not work well when comparing disaster data. Unfortunately, there is often no gold standard for disaster data because the data is collected so quickly and changes so frequently. In addition, many types of VGI data collected during the disaster were not collected by any authoritative source so there is simply nothing to compare them to. With these limitations in mind, this project attempts to compare the locations of internally displaced persons camps created by volunteers using the Ushahidi platform to locations created by trained remote sensors from various agencies. This comparison will provide us with a general sense of positional accuracy, temporal accuracy, and completeness.

This project required the use of several different data types. A combination of VGI vector data, authoritative vector data, remote sensing data, and authoritative maps were used to conduct the analysis. A description of each data type is below.

The original Ushahidi data set was downloaded from <http://www.haiti.ushahidi.com> in Spring of 2012. The original data set consisted of a CSV with fields for a unique identifier, title of record, record description, date/time stamp, location, category, latitude, longitude, verified status, and approved status. The original CSV consisted of approximately 4,600 records but a subset of those from the first two weeks following the earthquake were used for this project to better coincide with the aerial data that was available for the disaster zone. As part of a different project, the original data was recategorized and only those records that were a category 8a – Internally Displaced Persons Camp were used. This resulted in a sample size of 185 points.

The remote sensing data primarily covered Port-Au-Prince so an administrative boundary file for Haiti was used to select a regional boundary for Port-Au-Prince. This administrative boundary covered an area of 24,301,900 square meters. A variety of other vector data was downloaded to produce maps for the area of interest including transportation layers and hydrography layers.

The remote sensing data was from two different sources. Digital Globe and GeoEye provided data from the GeoEye-1 satellite for Port-Au-Prince captured on January 13, 2010, one day after the earthquake. The World Bank paid for aerial imagery to be collected and converted into a mosaiced orthophoto. Imagery was collected from January 15-28, 2010 but no information about the platform or sensor

was available. Satellite imagery was also collected from the Quickbird satellite on September 22, 2009 for temporal comparisons. A table summarizing the three remote sensing products is below.

Platform/Sensor	Spatial Resolution	Radiometric Resolution	Temporal Resolution	Spectral
Quickbird	0.64 x 0.64 meters	8 bit	September 22, 2009	Panchromatic
GeoEye-1	0.5 x 0.5 meters	8 bit	January 13, 2010	True Color RGB
Aerial Imagery	0.3 x 0.3 meters	8 bit	January 15-28, 2010	True Color RGB

To use for comparisons, several georectified maps were also acquired for the project. These maps are in the proprietary GeoPDF format which enables them to be added to ArcGIS and overlaid with traditional GIS data. A summary table for the four different maps used for this project are below.

Filename	Image Acquisition Date	Map Publication Date	Image Source	Analysis Source
F3A04278 B9989ED 1C12576B 00036CD 87- map.pdf	20100113	Unknown	GeoEye	UN Cartographic Section, G-Mosaic Project Partners, German Aerospace Centre
PDC_Data iled_Dam age_Map Compress ed.pdf	20100113	20100116	GeoEye	Pacific Disaster Center
UNOSAT_ HTI_EQ20 10_IDP_P P_v2_HR. pdf	20100113	20100115	GeoEye	UN Institute for Training and Research (UNITAR), UNOSAT
UNOSAT_ HTI_EQ20 10_IDPov erview_v 2_HR.pdf	20100113	20100117	GeoEye	UN Institute for Training and Research (UNITAR), UNOSAT

Due to the large area of the initial administrative boundary, a fishnet consisting of 500 meter x 500 meter grids cells was created in a GIS software platform that completely covered the study area. Each cell was given a unique and a random number generator was used to select 13 cells which were entirely in the study area, did not overlap water, and were primarily urban. A table summarizing the study area and grid cells is below.

Total Study Area Size (m ²)	24,301,900
Total Study Area Size (m ²) - only urban areas	21,432,200
Random Grid Selection Area (m ²)	3,250,000
% of Total Study Area Covered	13.37
% of Total Study Area - Only Urban Covered	15.16

The number of IDP camps in each of the four GeoPDF maps was calculated by heads up digitizing the images. Those points were then spatially joined to the randomly selected grid cells to generate a total number of IDP camps per cell. The same process was repeated for the Ushahidi data set. Finally, polygon shapefiles were generated for IDP camp boundaries visible in the GeoEye satellite imagery and World Bank commissioned aerial imagery. In order to distinguish between the rubble of collapsed buildings and IDP camps the Quickbird imagery from 2009 was compared to both the GeoEye and orthophoto imagery. To determine the distance from between the polygon IDP camp boundaries and the nearest point in the four map files and the Ushahidi data set, the ArcGIS Near tool was used.

A series of maps were created to highlight the study area, the randomly selected grid cells, and the locations of various point and polygon shapefiles.

Results

The results for this study can be broken down into three categories: completeness, positional accuracy, and temporal accuracy. A brief section for each category is below.

Completeness

There is no way to truly measure completeness for the data generated about internally displaced people camps that resulted from the January 12, 2010 earthquake in Haiti. The best we can do is compare the various data sets that were created to get a sense of how many camps were in each of the grid cells within the study area.

The VGI Ushahidi data set contains approximately the same number of IDP points within the total study area as the two UNOSAT maps and the UN map and significantly more than the PDC map. The total number of points within the grid cells is higher for the Ushahidi dataset but not significantly more than the UNOSAT and UN data. All data sets except the PDC data contain the same number of grid cells with points. It should be noted that this is not a fair comparison for completeness between the data sets however. The Ushahidi data set is from January 13 – 27 while the UNOSAT, PDC, and UN data are all based off of satellite imagery acquired on January 13. Because of that reason we would expect for the Ushahidi data set to contain more points than the other four maps. We could therefore assume that if the Ushahidi data set contains fewer points than the other four map data sets then it is likely incomplete.

Data Set	Total Points in Study Area	Total Points in Grid Cells	Cells with Points
Ushahidi	53	24	8
PDC	20	9	6
UNOSAT PAP	53	15	8
UNOSAT Overview	55	18	8
UN Cartographic	55	15	8

The following table compares the visual identification results conducted by the researcher on the two different remote sensing products captured after the disaster. While it appears that the Orthophoto contains many more IDP camp boundaries than the GeoEye satellite, again, this is not a fair comparison. The GeoEye satellite collected data on January 13 while the Orthophoto is a mosaic of images collected from January 15 – 28. We can make the same assumption based on completeness, however, and assume that if the Orthophoto contains fewer features than the GeoEye image then it is likely incomplete.

Data Set	Total Polygons in Grid Cells	Cells with Polygons
GeoEye-1	20	10
Orthophoto	42	9

Data for individual grid cells were also calculated and are presented in the table below. It is important to note that there was perfect agreement between all data sets for two of the cells (17 and 109). However, the PDC, UNOSAT, and UN data were all derived from the same GeoEye data so it is interesting to note that many of these cell counts disagree with each other and the counts produced with the raw GeoEye data digitized by the researcher.

Data Set	Cell 17	Cell 50	Cell 77	Cell 78	Cell 79	Cell 93	Cell 94	Cell 107	Cell 108	Cell 109	Cell 121	Cell 140	Cell 153
Ushahidi	0	0	2	1	3	1	12	0	2	0	2	0	1
PDC	0	1	0	2	1	0	2	1	0	0	0	2	0
UNOSAT PAP	0	2	1	3	2	1	2	0	2	0	0	2	0
UNOSAT Overview	0	3	1	3	3	1	3	0	2	0	0	2	0
UN Cartographic	0	3	1	2	0	1	2	0	2	0	0	3	1
GeoEye-1	0	3	1	3	2	1	2	1	2	0	0	4	1
Orthophoto	0	10	1	7	5	3	5	2	4	0	0	5	0

Positional Accuracy

Because the IDP camps do not contain uniquely identifying information there is no way to determine if any one point from Ushahidi or one of the maps corresponds to any one polygon from the two digitized images. However, we can use distance from polygon to nearest point as a proxy for positional accuracy. The table below details how many points were within set distances to camp boundary polygons for the Ushahidi data and the four map data sets.

GeoEye

Data Set	0 meters	<20 meters	20-100 meters	>100 meters
Ushahidi	0	2	6	12
PDC	6	3	4	7
UNOSAT PAP	7	6	6	1
UNOSAT Overview	5	7	7	1
UN Cartographic	8	7	3	2

Orthophoto

Data Set	0 meters	<20 meters	20-100 meters	>100 meters
Ushahidi	1	1	12	27
PDC	7	12	12	21
UNOSAT PAP	11	1	16	14
UNOSAT Overview	10	6	13	13
UN Cartographic	9	3	13	13

It is important to note that almost none of the Ushahidi points fall within a digitized camp boundary or even within 20 meters of the boundary. This may be significant considering how many of the points from the maps fall within a camp boundary or within 20 meters of one.

Temporal Accuracy

Just like with the other measures of quality above, it is impossible to know exactly how many IDP camps were operational at any given time. However, all of the data has a date stamp from either its data source or its publication. This information can be used to compare the different data sets and better understand the temporal components of the various data. The table below lists how many data points within the grid cells were available at any given time. The numbers are cumulative as you move horizontally across the table.

Data Set	1/13	1/15	1/16	1/17	1/27	1/28
Ushahidi	2	3	5	4	10	0
PDC	?	?	9	0	0	0
UNOSAT PAP	?	15	0	0	0	0
UNOSAT Overview	?	?	?	18	0	0
UN Cartographic	**The UN Cartographic map does not contain a published date					

It is important to note that the data set that was fastest to respond to the disaster and start providing data was the Ushahidi data. It was also the only data set that appeared to be updated throughout the disaster.

Discussion

Conclusions

The conclusions and discussion are broken into three sections below.

Completeness

It is somewhat difficult to draw any meaningful conclusions about completeness from this study without conducting some sort of statistical analysis. For instance, are the 24 points generated by the Ushahidi volunteers significantly more than the 18 generated by the UN remote sensing analysts? It is even more difficult when comparing the number of points within individual grid cells because they often differ by only one or two points. The one conclusion that can be drawn is that the Ushahidi dataset contains more points than the other data sets which is to be expected because it covers a larger temporal time frame.

Positional Accuracy

General trends in positional accuracy proxies for the four authoritative data sets and the one VGI data set are to be expected, however the extent may be surprising. The volunteers who were geocoding the Ushahidi data set did not have access to any of the tools that other geocoders use. For instance, there was no roads database and the maps that were originally available were generated by the Geospatial Intelligence Agency decades prior to the disaster. Therefore it is not surprising that none of the VGI points fall within the GeoEye polygonal boundaries and only one falls within the Orthophoto boundaries. It is also not surprising that many of the points fell further than 100 meters from the nearest polygonal boundary. The relatively higher positional accuracy of the four maps generated by relief agencies is expected since they were developed from the GeoEye data set. What may be surprising is that any points at all occurred more than 100 meters outside of the polygonal boundaries of their source. This may indicate a discrepancy in how IDP camps are identified by the various relief agencies and the researcher.

Temporal Accuracy

During disaster situations, conditions on the ground are constantly changing. As a result, data that is available more quickly and that is updated more often should be of the most use to relief agencies. In that instance, the data produced by the Ushahidi volunteers should have been the most useful as it was a dynamic and evolving data set. Because each record in the database contains a date and time stamp it is also possible to track conditions on the ground to the minute. Compare this to the map produced by the UN Cartographic Group that lacked any sort of published date at all. Responders and relief workers would then need to balance their trust of the United Nations with the uncertainty of the temporal accuracy of the data. Ultimately it is up to the responders to decide whether they want data of unknown quality now or data of assumed quality later.

Limitations

There are a number of limitations in this study. No proper statistical tests were conducted on the data so the researcher cannot say with any confidence that enough land area was covered or that enough data points were used. Nor can the researcher state that any of his findings have statistical significance. This study focused on only one part of the country and the results may have been different for regions that did not benefit from satellite coverage. Also, the study focused solely on an urban region and more rural regions may have different results. In addition, there were numerous other data sets that contained point data for suspected IDP camps that were not used in this data and they may have incorporated some different method to identify and locate IDP camps that was not used in this study. The individual images that were combined to create the orthophoto were taken over the course of two weeks. From a temporal standpoint it is difficult to draw conclusions without knowing the individual image collection dates. It was also challenging to determine what constituted an IDP camp. For instance, how many tents or tarps are needed to classify a group of people as a camp? Or should the definition rely on area covered instead? Many aid agencies were working within Haiti during this time and they may also have been hesitant to work from inside damaged buildings. Therefore it is possible that some of the camps identified were actually not camps at all but rather relief agencies that had set up temporary shelters. Finally, since there is no “gold standard” to compare the VGI data against, it really is difficult to state with any confidence the results of the above mentioned comparisons.

Future Research

There are many avenues for future research related to this project. Researchers should systematically address the limitations mentioned above related to statistical significance, area covered, types of area covered, and incorporating additional data sets. Other researchers could try expanding the particular topic of interest beyond IDP Camps. Ushahidi and traditional relief agencies generated data on blocked roads, damaged buildings, and food distribution centers. It would be interesting to learn if any of the trends found in this comparison translate to these other types of data. Finally, geospatial researchers may be able to use some form of geospatial statistics like kernel density, Local Moran's I, or GI* to develop models to test which data sets may best be used in the future for spatial decision making.

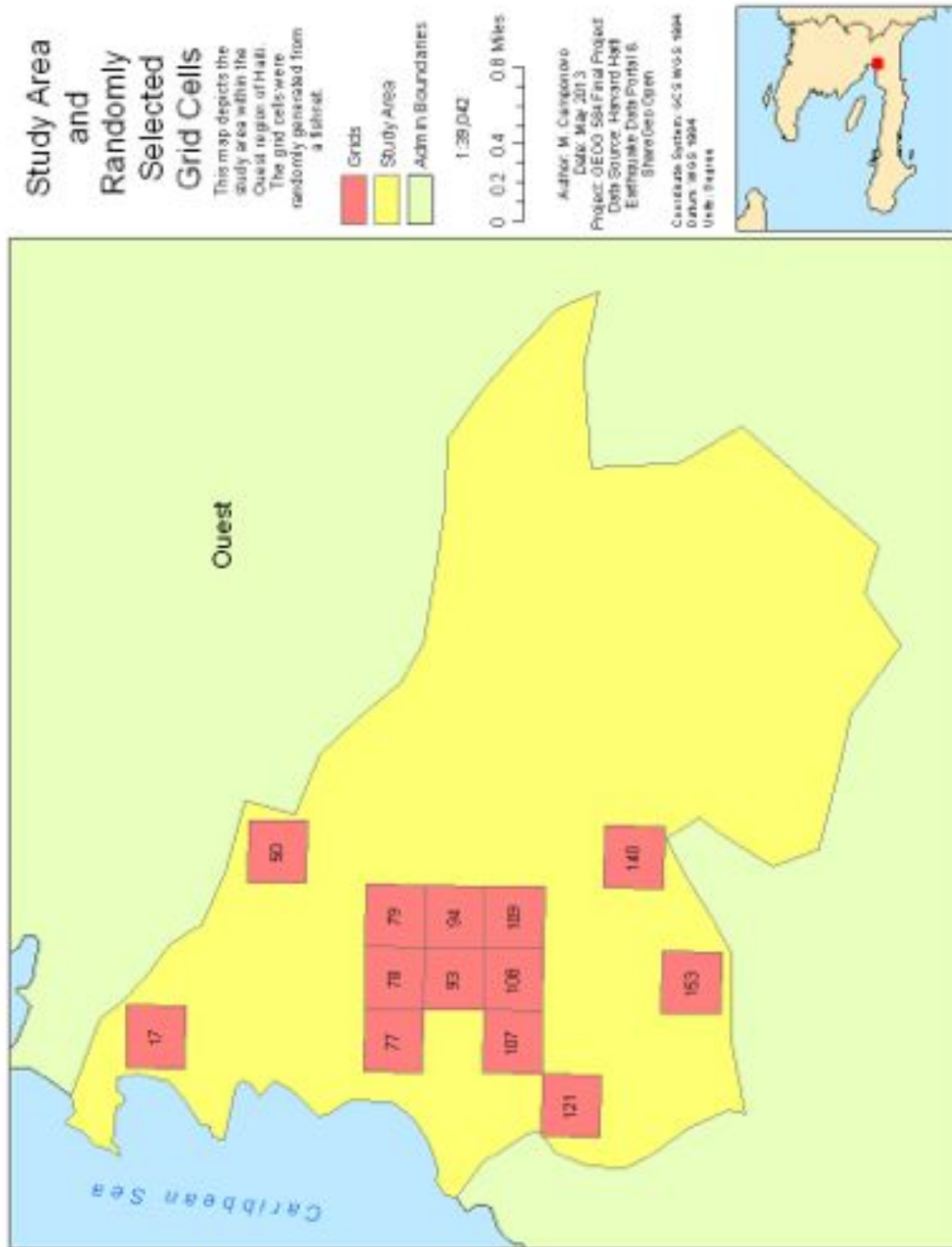
Appendices

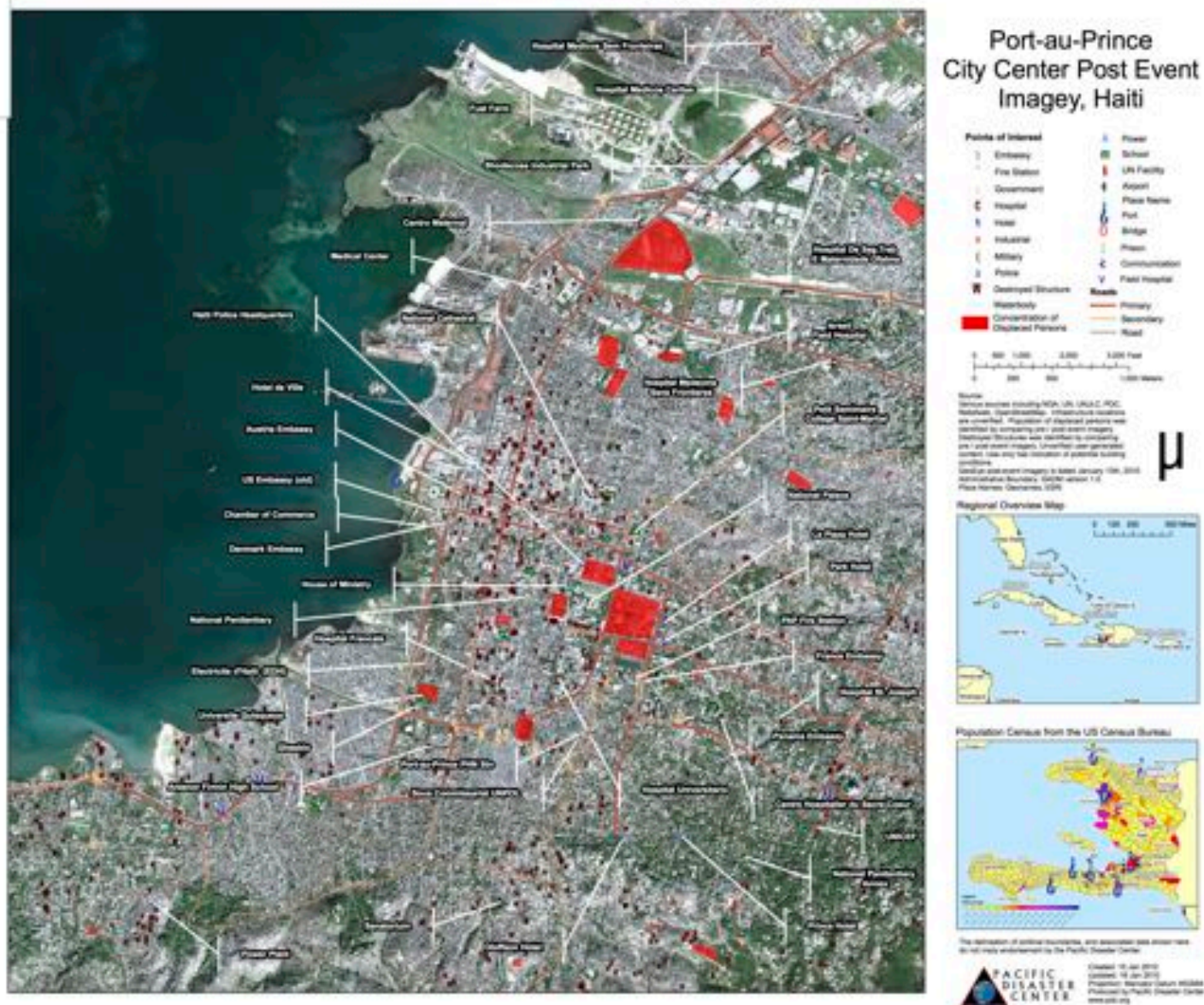
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Locator Map With Study Area Highlighted



Study Area with Grid Cells





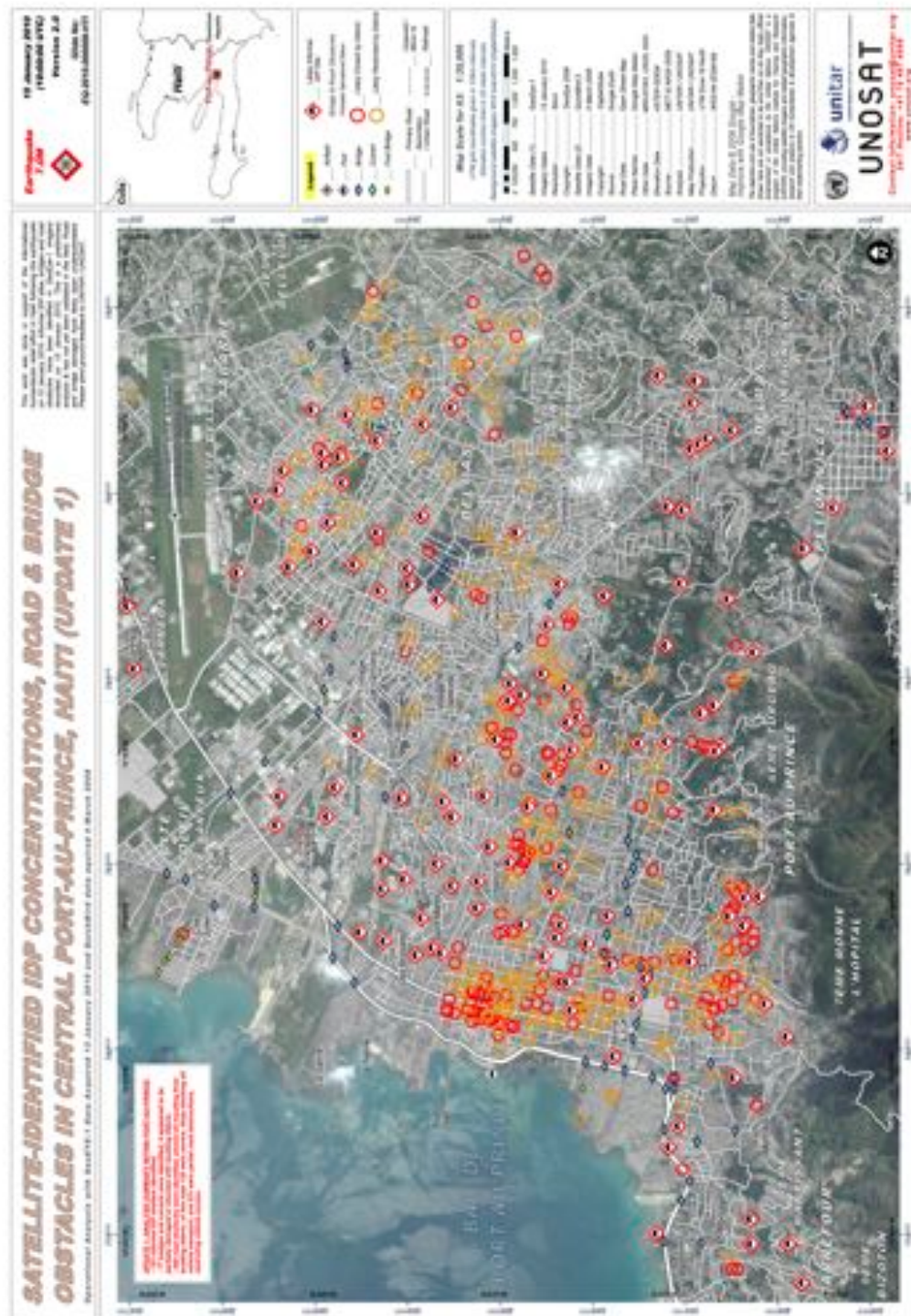






Image Comparison Map

Remote Sensing Imagery Comparison

This image was taken by the Quickbird satellite on September 22, 2009 prior to the earthquake. Notice the large open area in the center of the image.



This image was taken by the GeoEye-1 satellite on January 13, 2010, one day after the earthquake. Notice how the large open area in the center of the image has begun to fill with tarps.



This image was taken on behalf of the World Bank sometime between January 15 and January 26, 2010. Notice how the area that was previously empty has now completely filled with tarps and tents.

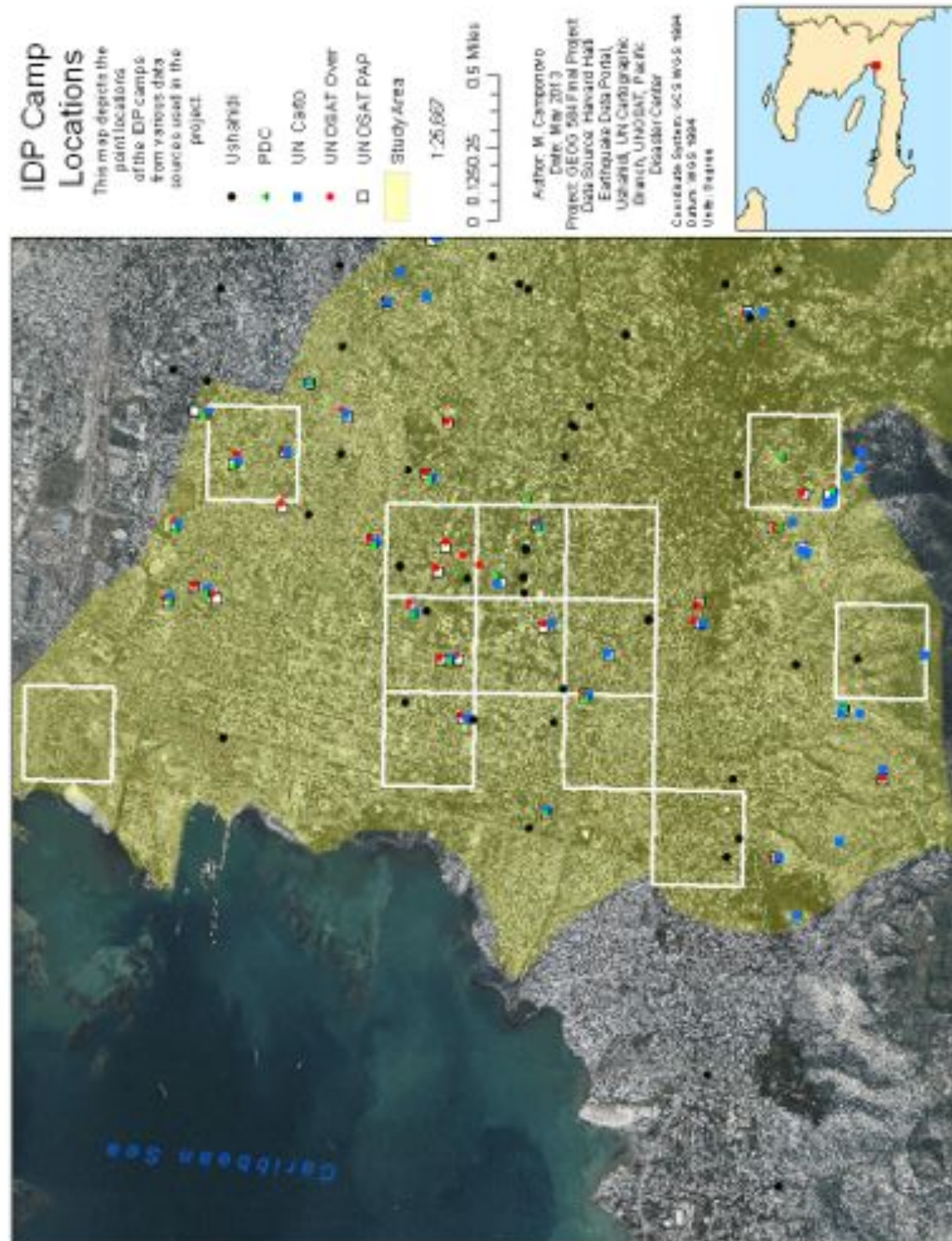


Author: M. Campion
Date: May 2013
Project: GEOG 594 Final Project
Data Source: Digital Globe and the World Bank

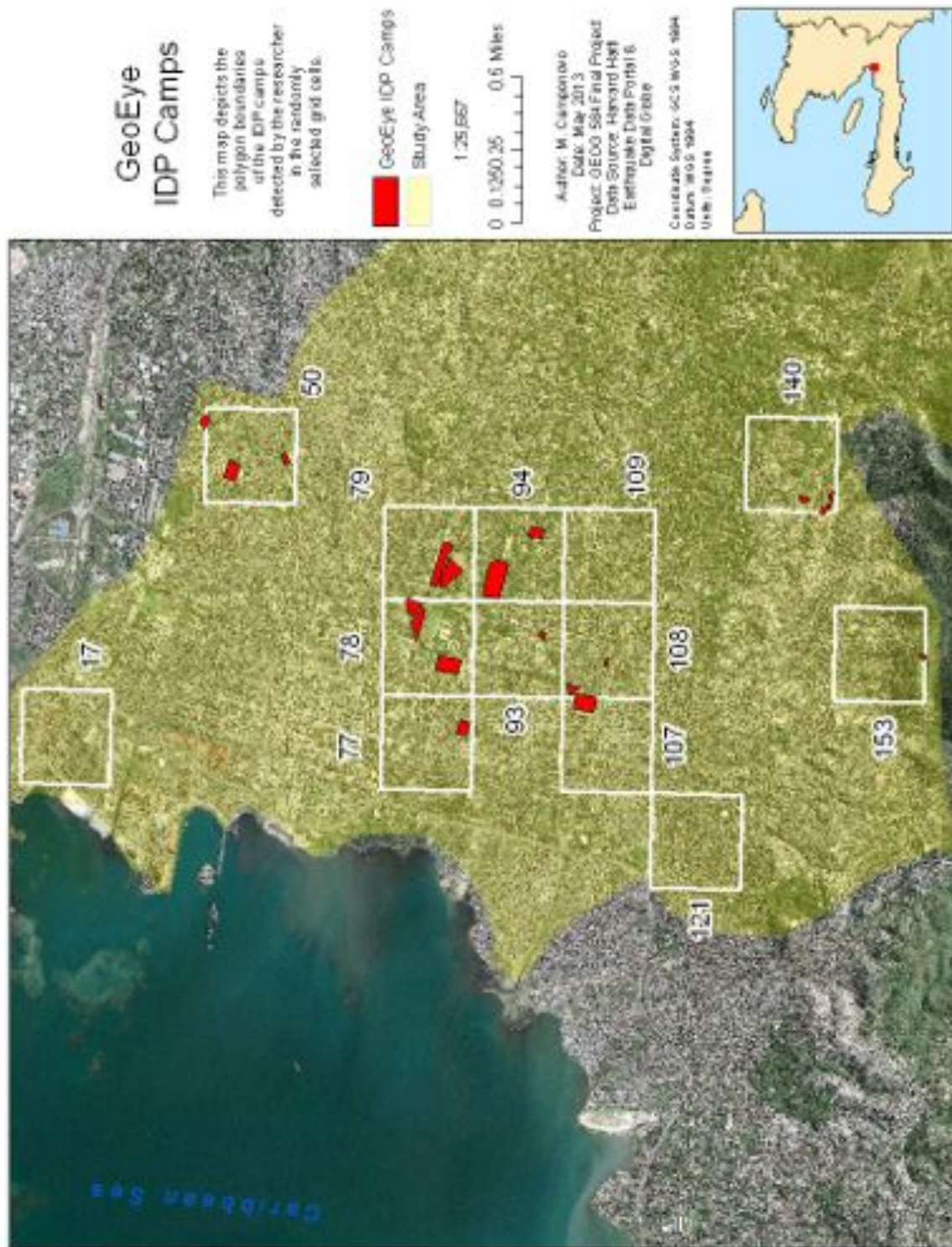
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1:3,140

Coordinate System: GCS WGS 1984
Datum: WGS 1984
Units: Degree

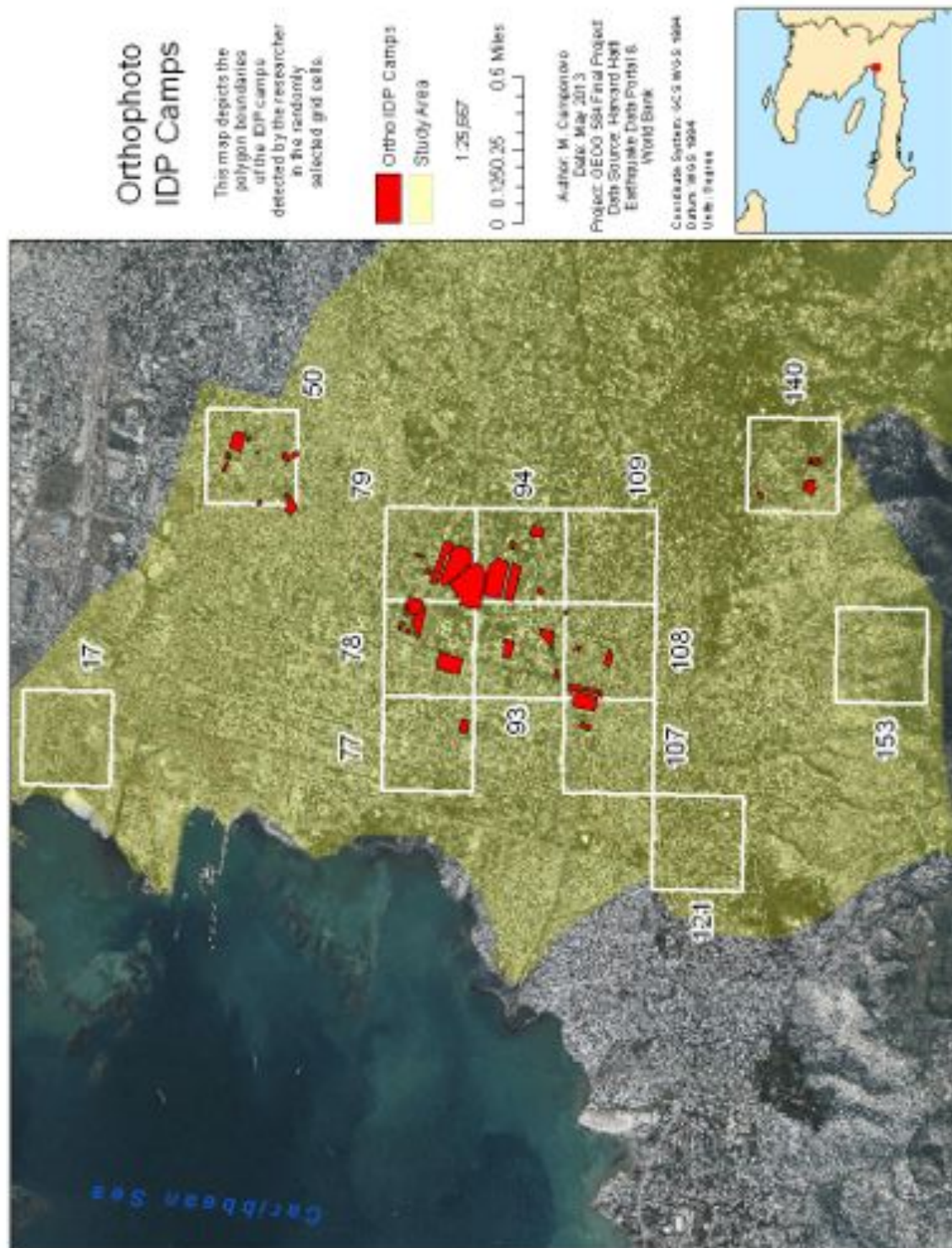
IDP Camp Point Locations



GeoEye IDP Camp Polygon Locations



Orthophoto IDP Camp Polygon Locations



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